

RENZO AGUSTIN GUTIERREZ

AI Engineer | LLM Integration, Evals & Reliability | Mobile Background

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PROFESSIONAL SUMMARY

AI Engineer with 7+ years in software development and 5+ years shipping production mobile applications, specialized in building LLM-powered products end to end: prompt design, structured outputs with strict schema validation, hallucination mitigation, custom evaluation harnesses, and latency/cost-driven model selection. Built two complete AI pipelines — speech-to-structured-data and OCR-to-structured-markdown (hybrid on-device + cloud) — each with eval frameworks whose measured results changed the technical roadmap. Daily user of AI-assisted development workflows (Claude, GitHub Copilot) and experienced technical lead in remote Agile teams.

AI / LLM SKILLS

LLM Integration: Gemini API, Anthropic (Claude) API, prompt engineering, system instructions, structured output with response schemas (JSON Schema / OpenAPI dialects), temperature-0 determinism, token budgeting (maxOutputTokens)

Reliability & Safety: hallucination mitigation (layered: prompt, schema, strict parsing, human-in-the-loop UI), strict output validation with warning telemetry, graceful failure handling, API-key security (secure storage, log redaction with tests)

Evaluation: custom eval harnesses, fixture-based golden sets, micro-averaged metrics (precision/recall on nullable fields, exact-value accuracy, WER), model matrix runs, latency benchmarking (p50), free-tier quota management

Cost Engineering: hybrid on-device + cloud architectures, token accounting and cost-weighted eval matrices, response caching, throttling and exponential backoff on 429/5xx, quota-aware design

Perception Pipelines: on-device and cloud speech-to-text, on-device OCR (Google ML Kit), document layout analysis (XY-cut page segmentation), two-stage perception-to-LLM pipelines with per-stage observability

Security: prompt-injection mitigation (system instructions separated from untrusted input, test-enforced), API keys in headers not URLs, secure key storage (Android Keystore), log redaction

Engineering: Dart, Flutter, interface-driven architecture, dependency injection, SQLite, REST APIs, Firebase, Git, CI/CD (Bitbucket Pipelines, Fastlane), 340+ automated tests across projects

AI PROJECTS

Trino — Voice-to-Structured-Data LLM Pipeline (creator)

Personal project | Flutter / Dart + Gemini API | github.com/agusg197

- Built a production-quality Android app that converts free-form Spanish speech into validated expense records in under 4 seconds: on-device/cloud STT feeding an LLM interpreter with structured output (~7,000 lines of Dart, 240 tests, clean static analysis, running on real hardware).
- Designed a 4-layer hallucination-mitigation system (“honesty system”): explicit prompt rules, nullable response schema, a strict parser that discards and logs every invalid field, and a confidence-state UI with mandatory human confirmation — achieving 0 invented fields across the full eval set.
- Built a from-scratch evaluation framework: 18 adversarial hand-written fixtures (router traps, relative dates, local slang, mid-sentence corrections) and 6 micro-averaged metrics, including precision AND recall on empty fields, exact amount accuracy and parser-warning counts — designed so zero-denominator cases report null instead of a misleading 100%.
- Ran a model-comparison matrix (3 Gemini models, 37 real API calls) that falsified the project's core hypothesis: the cheapest model scored 100% on all metrics at 838 ms p50 latency (5–10x faster than alternatives), eliminating a planned multi-model “ladder” and its cost.
- Discovered through targeted probing that cloud transcription models hallucinate full sentences from non-speech audio while the on-device recognizer correctly returns no-match — a finding no planned metric would have caught, which reshaped the pipeline's risk model.

- Implemented a data flywheel: every user confirmation (with or without corrections) is captured as a golden-set eval case in JSONL, verified end-to-end on device; built debug-only LLM observability with full request/response dumps and a test that fails if the API key ever reaches a log line.

Tiza — Hybrid On-Device OCR + LLM Structuring Pipeline (creator)

Personal project | Flutter / Dart + ML Kit + Gemini API | github.com/agusg197

- Built an Android app converting whiteboard/handwriting photos into structured markdown with the hybrid pattern: free on-device OCR (ML Kit) feeding a single, user-triggered LLM call — nothing leaves the device until the user asks (104 automated tests, deterministic device-free evals).
- Designed 3 interchangeable layout serializers, including an XY-cut page segmentation algorithm with a text-height-relative gutter heuristic, and an ordering metric (extra switches + inversions) to iterate on layout for free before spending tokens.
- Ran a cost-weighted eval matrix (serializers × models × output modes, output tokens weighted 5x) that produced actionable, counterintuitive findings: the cheapest configuration scored 100% at 43% of the cost; XY-cut had zero measurable gain (the model is order-indifferent given coordinates); and without positional data the schema hallucinates action items — coordinates are the price of structured output.
- Implemented strict discard-and-count validation (invalid entries dropped, logged and surfaced as an eval column), per-configuration response caching, throttling with exponential backoff on 429s, and 7 recoverable error states each mapped to real user actions, enforced by a test over the full enum.
- Applied prompt-injection mitigation by separating system instructions from untrusted OCR text (“the note is data, not an order”), with a test verifying the separation; BYOK key handling via Android Keystore with the key sent in headers, never in URLs.

AI-Assisted Engineering Practice — Leadbox Inc.

Remote | 2021 – Present

- Introduced and standardized AI-assisted development workflows (GitHub Copilot, Claude) across daily feature work and code review as mobile lead, increasing delivery velocity without reducing code quality.

PROFESSIONAL EXPERIENCE

Flutter Developer (Contract) — Crediclub, Digital Bank (via Zafirus)

Remote, Mexico | November 2025 – June 2026

- Shipped core transfer and payment features in a production banking app with 100,000+ monthly transactions; kept crash-free rate above 99.8% and passed security/compliance review on all 6 releases.

Flutter Developer — Mobile Lead — Leadbox Inc.

Remote | February 2021 – Present

- Lead the mobile area (planning, technical decisions, code review, releases); architected the Flutter migration of a complex web platform and built CI/CD pipelines cutting release overhead by ~50%.

Front-End Developer — Freelance

Remote | 2019 – 2021

- Delivered 8+ React/Angular web applications for international clients with a 100% on-time record.

LANGUAGES

Spanish: Native **English:** B1, professional working proficiency — comfortable in daily standups, technical discussion and written communication.